Class 11: AlphaFold

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Background

In this hands-on session we will utilize AlphaFold to predict protein structure from sequence (Jumper et al. 2021).

Without the aid of such approaches, it can take years of expensive laboratory work to determine the structure of just one protein. With AlphaFold we can now accurately compute a typical protein structure in as little as ten minutes.

The PDB database (the main repository of experimental structures) only has ~**250 thousand** structures (we saw this in the last lab). The main protein sequence database has over **200 million** sequences! Only 0.125% of known sequences have a known structure - this is called the "structure knowledge gap".

```
(250000 / 200000000) * 100
```

```
[1] 0.125
```

- Structures are much harder to determine than sequences
- They are expensive (on average ~$1 million each)
- They take on average 3-5 yeras to solve!

EBI AlphaFold Database

The EBI has a database of pre-computed AlphaFold (AF) models called AFDB. This is growing all the time and can be useful to check before running AF ourselves.

Running AlphaFold

We can download and run locally (on our own computers) but we need a GPU. Or we can use "cloud" computing to run this on someone else's computers.

We will use ColabFold < https://github.com/sokrypton/ColabFold >

We previously found there was no AFDB entry for our HIV sequence:

```
>HIV-Pr-Dimer
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF:PQITLWQRPLVTIKIGGQLK
EALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPT
PVNIIGRNLLTQIGCTLNF
```

Here we will use AlphaFold2_mmseqs2

```r
results_dir <- "hivpr_23119"
```

```r
# File names for all PDB models
pdb_files <- list.files(path=results_dir,
                        pattern="*pdb",
                        full.names = TRUE)

# Print our PDB files names
basename(pdb_files)
```

```
[1] "hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb"
[2] "hivpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb"
[3] "hivpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb"
[4] "hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb"
[5] "hivpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb"
```

```r
library(bio3d)

# Read all data from Models
#  and superpose/fit coords
pdbs <- pdbaln(pdb_files, fit=TRUE, exefile="msa")
```

```
Reading PDB files:
hivpr_23119/hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb
hivpr_23119/hivpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb
hivpr_23119/hivpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb
hivpr_23119/hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb
hivpr_23119/hivpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb
.....

Extracting sequences

pdb/seq: 1   name: hivpr_23119/hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4
pdb/seq: 2   name: hivpr_23119/hivpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1
pdb/seq: 3   name: hivpr_23119/hivpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5
pdb/seq: 4   name: hivpr_23119/hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2
pdb/seq: 5   name: hivpr_23119/hivpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3
```

```r
pdbs
```

```
                              1        .         .         .         .         50
[Truncated_Name:1]hivpr_2311  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:2]hivpr_2311  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:3]hivpr_2311  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:4]hivpr_2311  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:5]hivpr_2311  PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
                              **************************************************
                              1        .         .         .         .         50

                             51        .         .         .         .         100
[Truncated_Name:1]hivpr_2311  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:2]hivpr_2311  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:3]hivpr_2311  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:4]hivpr_2311  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:5]hivpr_2311  GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
                              **************************************************
                             51        .         .         .         .         100
```

```
                              101        .         .         .         .         150
[Truncated_Name:1]hivpr_2311   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:2]hivpr_2311   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:3]hivpr_2311   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:4]hivpr_2311   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:5]hivpr_2311   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
                               **************************************************
                              101        .         .         .         .         150

                              151        .         .         .         .       198
[Truncated_Name:1]hivpr_2311   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:2]hivpr_2311   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:3]hivpr_2311   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:4]hivpr_2311   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:5]hivpr_2311   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
                               **************************************************
                              151        .         .         .         .       198

Call:
  pdbaln(files = pdb_files, fit = TRUE, exefile = "msa")

Class:
  pdbs, fasta

Alignment dimensions:
  5 sequence rows; 198 position columns (198 non-gap, 0 gap)

+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

```r
rd <- rmsd(pdbs, fit=T)
```

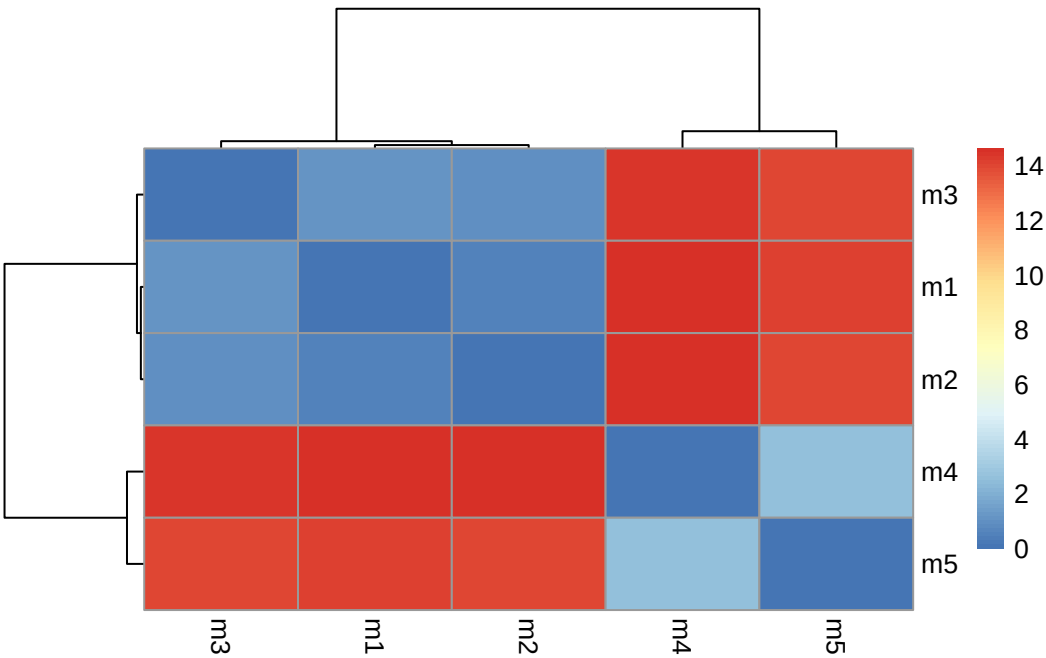
```
Warning in rmsd(pdbs, fit = T): No indices provided, using the 198 non NA positions
```

```r
range(rd)
```

```
[1]  0.000 14.628
```

```r
library(pheatmap)
```

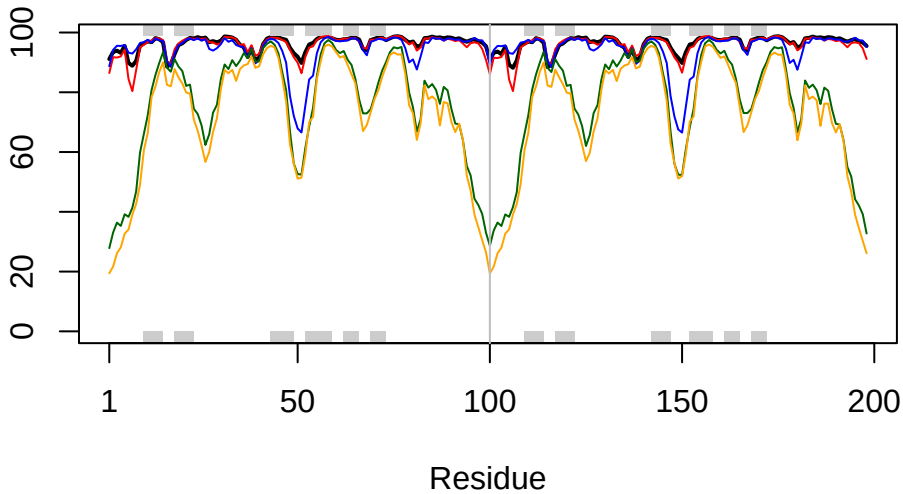
```r
colnames(rd) <- paste0("m",1:5)
rownames(rd) <- paste0("m",1:5)
pheatmap(rd)
```




```r
# Read a reference PDB structure
pdb <- read.pdb("1hsg")
```

```
  Note: Accessing on-line PDB file
```

```r
plotb3(pdbs$b[1,], typ="l", lwd=2, sse=pdb)
points(pdbs$b[2,], typ="l", col="red")
points(pdbs$b[3,], typ="l", col="blue")
points(pdbs$b[4,], typ="l", col="darkgreen")
points(pdbs$b[5,], typ="l", col="orange")
abline(v=100, col="gray")
```




```r
core <- core.find(pdbs)
```

```
core size 197 of 198  vol = 8788.529
core size 196 of 198  vol = 3258.252
core size 195 of 198  vol = 1641.096
core size 194 of 198  vol = 1498.73
core size 193 of 198  vol = 1369.427
core size 192 of 198  vol = 1313.823
core size 191 of 198  vol = 1228.996
core size 190 of 198  vol = 1186.704
core size 189 of 198  vol = 1146.395
core size 188 of 198  vol = 1102.479
core size 187 of 198  vol = 1065.499
core size 186 of 198  vol = 1024.388
core size 185 of 198  vol = 990.444
core size 184 of 198  vol = 959.404
core size 183 of 198  vol = 896.555
core size 182 of 198  vol = 862.291
core size 181 of 198  vol = 813.796
core size 180 of 198  vol = 778.382
core size 179 of 198  vol = 752.381
core size 178 of 198  vol = 727.481
```

```
core size 177 of 198  vol = 701.894
core size 176 of 198  vol = 665.126
core size 175 of 198  vol = 645.103
core size 174 of 198  vol = 611.674
core size 173 of 198  vol = 592.081
core size 172 of 198  vol = 574.373
core size 171 of 198  vol = 557.314
core size 170 of 198  vol = 540.852
core size 169 of 198  vol = 521.716
core size 168 of 198  vol = 504.275
core size 167 of 198  vol = 486.79
core size 166 of 198  vol = 467.6
core size 165 of 198  vol = 453.621
core size 164 of 198  vol = 438.703
core size 163 of 198  vol = 416.355
core size 162 of 198  vol = 403.203
core size 161 of 198  vol = 389.116
core size 160 of 198  vol = 380.198
core size 159 of 198  vol = 368.51
core size 158 of 198  vol = 354.354
core size 157 of 198  vol = 345.969
core size 156 of 198  vol = 333.055
core size 155 of 198  vol = 323.291
core size 154 of 198  vol = 310.454
core size 153 of 198  vol = 298.867
core size 152 of 198  vol = 287.369
core size 151 of 198  vol = 274.702
core size 150 of 198  vol = 265.237
core size 149 of 198  vol = 253.87
core size 148 of 198  vol = 242.239
core size 147 of 198  vol = 231.934
core size 146 of 198  vol = 221.935
core size 145 of 198  vol = 212.751
core size 144 of 198  vol = 202.011
core size 143 of 198  vol = 190.404
core size 142 of 198  vol = 178.064
core size 141 of 198  vol = 166.007
core size 140 of 198  vol = 154.581
core size 139 of 198  vol = 148.005
core size 138 of 198  vol = 142.521
core size 137 of 198  vol = 137.976
core size 136 of 198  vol = 133.424
core size 135 of 198  vol = 127.466
```

```
core size 134 of 198  vol = 122.719
core size 133 of 198  vol = 116.921
core size 132 of 198  vol = 111.806
core size 131 of 198  vol = 107.066
core size 130 of 198  vol = 102.136
core size 129 of 198  vol = 95.713
core size 128 of 198  vol = 91.571
core size 127 of 198  vol = 87.169
core size 126 of 198  vol = 83.202
core size 125 of 198  vol = 79.384
core size 124 of 198  vol = 75.619
core size 123 of 198  vol = 72.22
core size 122 of 198  vol = 68.808
core size 121 of 198  vol = 65.053
core size 120 of 198  vol = 59.938
core size 119 of 198  vol = 56.462
core size 118 of 198  vol = 50.965
core size 117 of 198  vol = 47.711
core size 116 of 198  vol = 44.559
core size 115 of 198  vol = 42.227
core size 114 of 198  vol = 38.533
core size 113 of 198  vol = 35.532
core size 112 of 198  vol = 32.536
core size 111 of 198  vol = 30.047
core size 110 of 198  vol = 27.602
core size 109 of 198  vol = 25.46
core size 108 of 198  vol = 23.828
core size 107 of 198  vol = 21.844
core size 106 of 198  vol = 21.094
core size 105 of 198  vol = 18.919
core size 104 of 198  vol = 18.031
core size 103 of 198  vol = 16.952
core size 102 of 198  vol = 15.508
core size 101 of 198  vol = 13.766
core size 100 of 198  vol = 13.412
core size 99 of 198  vol = 12.055
core size 98 of 198  vol = 10.539
core size 97 of 198  vol = 9.331
core size 96 of 198  vol = 7.81
core size 95 of 198  vol = 6.417
core size 94 of 198  vol = 5.507
core size 93 of 198  vol = 4.463
core size 92 of 198  vol = 3.688
```

```
 core size 91 of 198  vol = 2.703
 core size 90 of 198  vol = 2.106
 core size 89 of 198  vol = 1.512
 core size 88 of 198  vol = 1.06
 core size 87 of 198  vol = 0.777
 core size 86 of 198  vol = 0.641
 core size 85 of 198  vol = 0.524
 core size 84 of 198  vol = 0.473
 FINISHED: Min vol ( 0.5 ) reached
```

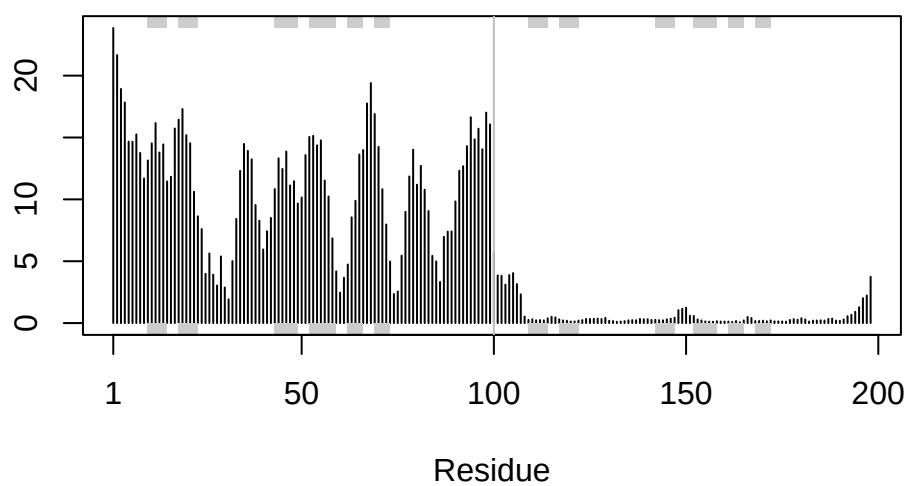
```r
core.inds <- print(core, vol=0.5)
```

```
# 85 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1     8  48     41
2    52  95     44
```

```r
xyz <- pdbfit(pdbs, core.inds, outpath="corefit_structures")
```

```r
rf <- rmsf(xyz)

plotb3(rf, sse=pdb)
abline(v=100, col="gray", ylab="RMSF")
```



Predicted Alignment Error for domains

```r
library(jsonlite)

# Listing of all PAE JSON files
pae_files <- list.files(path=results_dir,
                        pattern=".*model.*\\.json",
                        full.names = TRUE)
```

```r
pae1 <- read_json(pae_files[1],simplifyVector = TRUE)
pae5 <- read_json(pae_files[5],simplifyVector = TRUE)

attributes(pae1)
```

```
$names
[1] "plddt"   "max_pae" "pae"     "ptm"     "iptm"
```

```r
# Per-residue pLDDT scores
#  same as B-factor of PDB..
head(pae1$plddt)
```

```
[1] 91.12 93.50 93.94 93.19 95.50 89.81
```

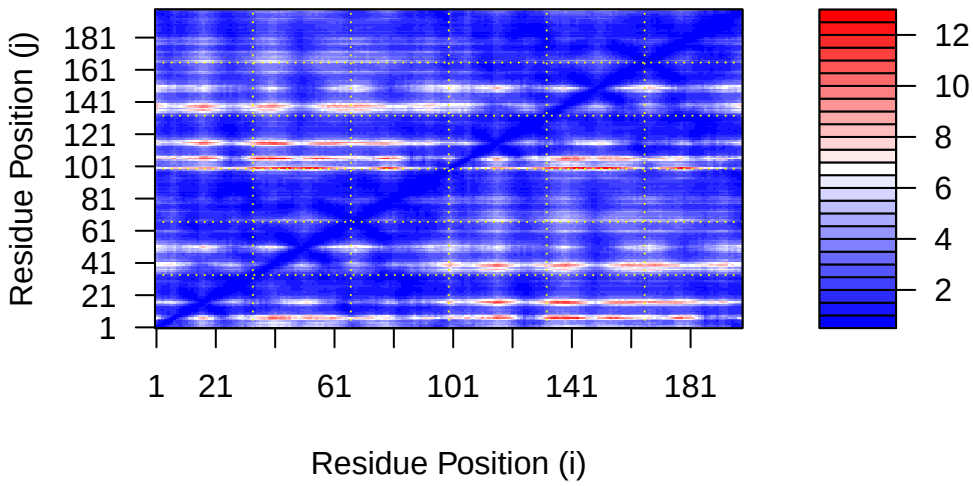
```r
pae1$max_pae
```

```
[1] 12.70312
```

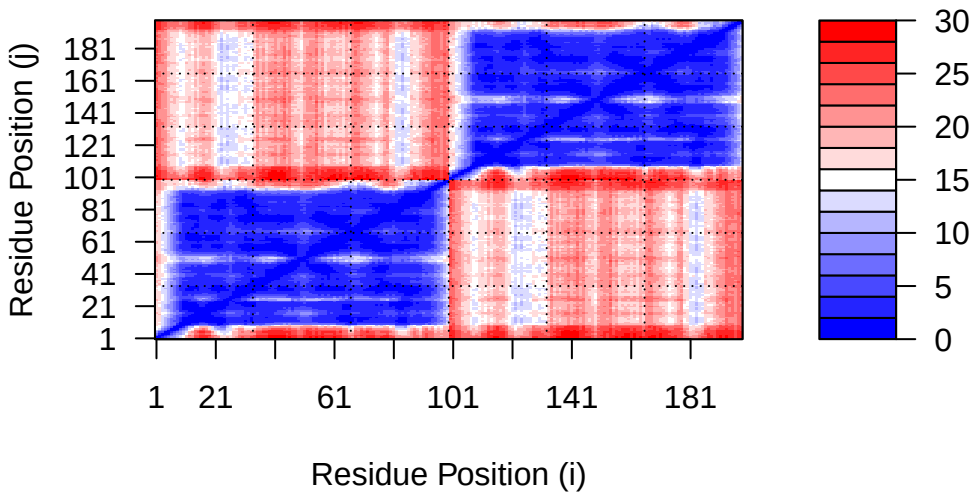
```r
pae5$max_pae
```

```
[1] 29.76562
```

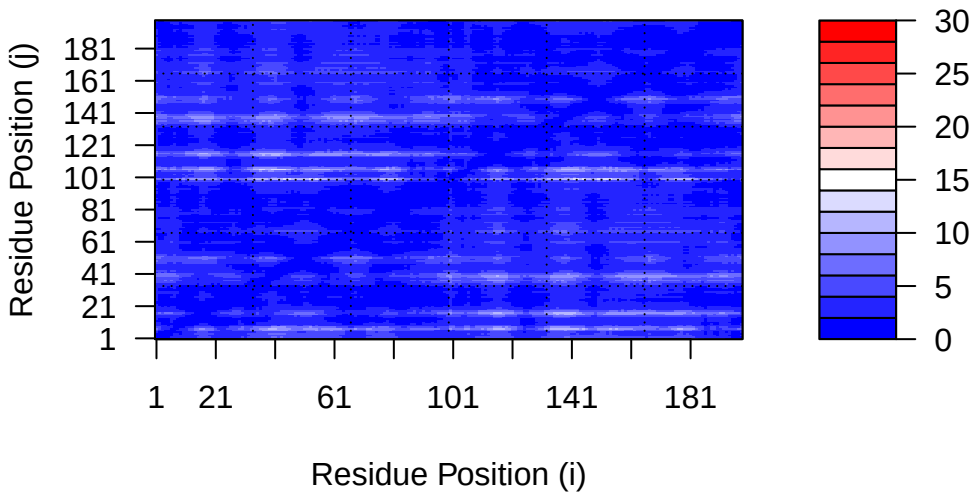
```r
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)")
```




```r
plot.dmat(pae5$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```




```r
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```



Residue conservation from alignment file

```
aln_file <- list.files(path=results_dir,
                       pattern=".a3m$",
                        full.names = TRUE)
aln_file
```

```
[1] "hivpr_23119/hivpr_23119.a3m"
```

```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

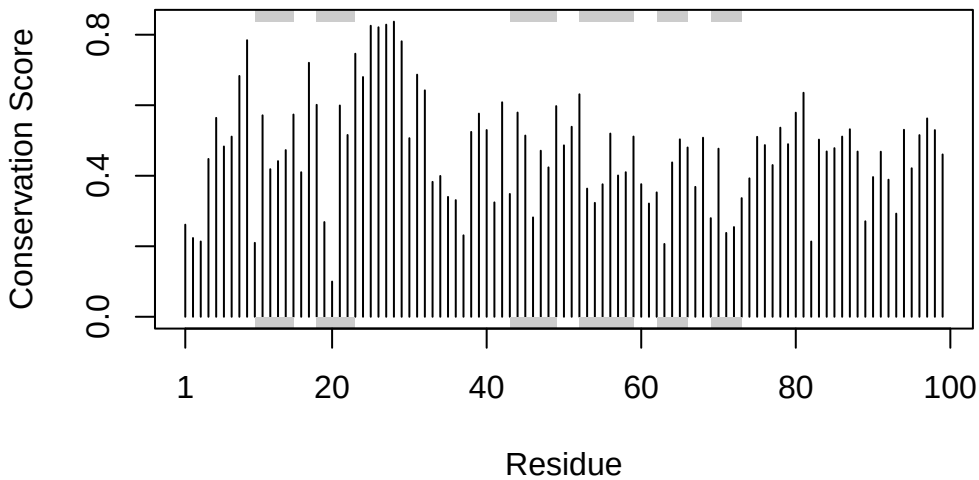
```
[1] " ** Duplicated sequence id's: 101 **"
[2] " ** Duplicated sequence id's: 101 **"
```

```
dim(aln$ali)
```

```
[1] 5397  132
```

```r
sim <- conserv(aln)
```

```r
plotb3(sim[1:99], sse=trim.pdb(pdb, chain="A"),
       ylab="Conservation Score")
```




```r
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
  [1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-" "-"
 [37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```

```r
m1.pdb <- read.pdb(pdb_files[1])
occ <- vec2resno(c(sim[1:99], sim[1:99]), m1.pdb$atom$resno)
write.pdb(m1.pdb, o=occ, file="m1_conserv.pdb")
```